# Quan Xiao | Curriculum Vitae

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## EDUCATION

**Cornell University** — **New York, NY**
*Ph.D. in Electrical and Computer Engineering (GPA: 4.3/4.3)* — *Aug. 2025 - present*
Advisor: Tianyi Chen

**Rensselaer Polytechnic Institute** — **Troy, NY**
*Ph.D. of Computer and Systems Engineering (GPA: 4.0/4.0)* — *Sep. 2021 - (Transferred) Aug. 2025*
Advisor: Tianyi Chen

**University of Science and Technology of China** — **Hefei, Anhui, China**
*Bachelor of Statistics* — *Sep. 2016 - July 2020*

## RESEARCH INTEREST

My research focuses on optimization and machine learning, particularly **bilevel and multi-objective optimization**, to develop 1) **efficient methods for generative and agentic AI that coordinate interacting components and balance competing objectives**, including LLM fine-tuning, retrieval-augmented reasoning, tool use, memory management, and trade-offs between training efficiency and exploration; and 2) **hardware–algorithm co-design methods for energy-efficient AI training and deployment**, explicitly accounting for physical constraints and device nonidealities in **analog in-memory computing (AIMC)**. More broadly, I aim to bridge theory and practice by designing scalable and principled methods for safe, reliable, efficient, and hardware-aware AI systems.

## HONORS, AWARDS, AND RESEARCH FUNDING

Google Cloud Research Credits Program ($1,000 in API credits) — *2026 - 2027*
NSF ACCESS Award CIS261228: Taming Competing Objectives in Tool-Using LLM Agents — *2026 - 2027*
*Co-Principal Investigator; total award: 750,000 ACCESS GPU credits*
Lambda Research Grant ($2,000 in cloud credits) — *2026*
ICML Gold Reviewer — *2026*
Travel Award for NYC Ops Day — *2026*
Cisco Research Award (*Co-Proposal Writer; total award: $75,000*) — *2025*
NeurIPS Top Reviewer — *2025*
RPI Institutional Nominee for Google PhD Fellowship (one of four university-wide) — *2024*
IEEE Signal Processing Society Scholarship — *2024*
Belsky Award for Computational Sciences and Engineering in RPI — *2023*
Neural Information Processing Systems Scholar Award — *2023*
International Conference on Machine Learning Participation Grant — *2022*
Successful Participant in Mathematical Contest In Modeling — *2019*
Excellent student scholarship of the University of Science and Technology of China — *2017 - 2019*

## SELECTED PUBLICATIONS AND PREPRINTS

*For a complete list of my publications, please see my Google Scholar.*

### Optimization Theory

[12] Liuyuan Jiang, **Quan Xiao**, Lisha Chen, and Tianyi Chen. Beyond value functions: Single-loop bilevel optimization under flatness conditions. *Proc. of Neural Information Processing Systems*, 2025.

[10] Lisha Chen, **Quan Xiao**, Ellen Hidemi Fukuda, Xinyi Chen, Kun Yuan, Tianyi Chen. Efficient Multi-Objective Learning under Preference Guidance: A First-Order Penalty Approach. *Proc. of International Conference on Machine Learning*, 2025. **(Spotlight, top 2.6%)**

[9] **Quan Xiao**, and Tianyi Chen. Unlocking Global Optimality in Bilevel Optimization: A Pilot Study. *Proc. of International Conference on Learning Representations*, 2025.

[8] Han Shen, **Quan Xiao**, and Tianyi Chen. On Penalty-based Bilevel Gradient Descent Method. *Mathematical Programming (Series A)*, 2024.

[7] Liuyuan Jiang, **Quan Xiao**, Victor M Tenorio, Fernando Real-Rojas, Antonio Marques, Tianyi Chen. A Primal-Dual-Assisted Penalty Approach to Bilevel Optimization with Coupled Constraints. *Proc. of Neural Information Processing Systems*, 2024

[5] **Quan Xiao**, Songtao Lu, and Tianyi Chen. A Generalized Alternating Method for Bilevel Learning under the Polyak-Łojasiewicz Condition. *Proc. of Neural Information Processing Systems*, 2023.

[4] **Quan Xiao**, Qing Ling, and Tianyi Chen. Lazy Queries Can Reduce Variance in Zeroth-Order Optimization. *IEEE Transactions on Signal Processing*, 2023, vol. 71, pp. 3695–3709.

[3] **Quan Xiao**, Han Shen, Wotao Yin and Tianyi Chen. Alternating Projected SGD for Equality-constrained Bilevel Optimization. *Proc. of International Conference on Artificial Intelligence and Statistics*, 2023.

[1] Tianyi Chen, Yuejiao Sun, **Quan Xiao**, and Wotao Yin. A Single-Timescale method for stochastic bilevel optimization. *International Conference on Artificial Intelligence and Statistics*, 2022. **(Oral, top 2%)**

### Generative, agentic AI and other ML applications

[18] **Quan Xiao**, Mingda Liu, Gaowen Liu, Katsuki Fujisawa, and Tianyi Chen. BRIDGE: Bilevel Retrieval-Credit-Aware Agentic Reinforcement Learning. arXiv preprint arXiv:2609.36505, 2026.

[17] Heshan Fernando, **Quan Xiao**, Yan Xin, and Tianyi Chen. ARMOR: Adaptive Retriever Optimization for Low-Resource Telecom Question Answering. arXiv:2606.29706, 2026.

[16] **Quan Xiao**, Yutong Xuan, Gaowen Liu, Ramana Rao Kompella, and Tianyi Chen. Bilevel Data Curation for LLM Fine-tuning: Offline Selection and Online Self-Refining Generation. arXiv:2511.21056, 2026.

[11] **Quan Xiao**, Hui Yuan, A F M Saif, Gaowen Liu, Ramana Rao Kompella, Mengdi Wang, Tianyi Chen. A First-order Generative Bilevel Optimization Framework for Diffusion Models. *Proc. of International Conference on Machine Learning*, 2025.

[6] Jiajia Yu, **Quan Xiao**, Tianyi Chen, and Rongjie Lai. A Bilevel Optimization Method for Inverse Mean-Field Games. *Inverse Problems*, 2024.

[2] *Momin Abbas, ***Quan Xiao**, *Lisha Chen, Pin-Yu Chen and Tianyi Chen. Sharp-MAML: Sharpness-Aware Model-Agnostic Meta Learning. *Proc. of International Conference on Machine Learning*, 2022.

### Analog In-Memory Computing

[20] Zhaoxian Wu, **Quan Xiao**, Tayfun Gokmen, and Tianyi Chen. Optimization under Persistent State-Dependent Bias: Gradient-based Method and Complexity Analysis. arXiv:2607.26032, 2026.

[19] **Quan Xiao**, Jindan Li, Zhaoxian Wu, Tayfun Gokmen, and Tianyi Chen. Breaking the Asymmetry Barrier in Analog In-Memory Training: A Signal Processing Solution. *Asilomar Conference on Signals, Systems, and Computers*, 2026.

[15] ***Quan Xiao**, *Jindan Li, Zhaoxian Wu, Tayfun Gokmen, and Tianyi Chen. Dynamic Symmetric Point

Tracking: Tackling Non-ideal Reference in Analog In-memory Training. *Proc. of International Conference on Machine Learning*, 2026.

[14] Zhaoxian Wu, **Quan Xiao**, Tayfun Gokmen, Hsinyu Tsai, Kaoutar El Maghraoui, Tianyi Chen. On the Convergence Theory of Pipeline Gradient-based Analog In-memory Training. *IEEE Journal on Selected Areas in Information Theory*, 2026.

[13] Zhaoxian Wu, **Quan Xiao**, Tayfun Gokmen, Omobayode Fagbohungbe, and Tianyi Chen. Analog In-memory Training on General Non-ideal Resistive Elements: Understanding the Impact of Response Functions. *Proc. of Neural Information Processing Systems*, 2025. **(Oral, top 0.3%)**

## INVITED TALKS

- **University of California, Santa Cruz**, ECE Special Seminar. Oct. 2026
  *Bilevel Optimization for Agentic AI: From Theory to Retrieval-augmented Agentic Reinforcement Learning.*
- **NYC Operations (Ops) Day**, New York, NY. Apr. 2026
  *Unlocking Nested Learning for AI: A Case Study on Data Selection for LLM.*
- **Information Theory and Applications (ITA) Workshop**, San Diego, CA. Feb. 2026
  *Unlocking Nested Learning for AI: A Case Study on Data Selection for LLM.*
- **Conference on Parsimony and Learning (CPAL)**, Stanford, CA. Mar. 2025
  *Global Convergence in Bilevel Optimization: A Pilot Study.*
- **Asilomar Conference on Signals, Systems, and Computers**, Pacific Grove, CA. Nov. 2024
  *Global Convergence in Bilevel Optimization: A Pilot Study.*
- **Asilomar Conference on Signals, Systems, and Computers**, Pacific Grove, CA. Nov. 2023
  *Efficient Methods for Bilevel Optimization with Polyak-Lojasiewicz Lower-level Problems.*
- **SIAM Conference on Optimization (OP23)**, Seattle, WA. June 2023
  *Efficient Methods for Bilevel Optimization with Polyak-Lojasiewicz Lower-level Problems.*

## TEACHING EXPERIENCE

- **Teaching Assistant, ECE 5110: Random Signals in Communications and Signal Processing**
  Cornell University, graduate level Spring 2026
- **Guest Lecturer, ECSE 4964/6964: Distributed Optimization & Machine Learning**
  Rensselaer Polytechnic Institute, graduate level Spring 2025
- **Guest Lecturer, ECSE 2500: Engineering Probability**
  Rensselaer Polytechnic Institute, undergraduate level Fall 2024

## INTERNSHIP

- **IBM Thomas J. Watson Research Center**, Yorktown, NY.
  Research intern with mentor: Dr. Debarun Bhattacharjya. May. 2024 - Aug. 2024
  **Project:** Uncertainty quantification and selective generation in large language model.
- **IBM Thomas J. Watson Research Center**, Yorktown, NY.
  Research intern with mentor: Dr. Songtao Lu. May. 2022 - Aug. 2022
  **Project:** Efficient method for bilevel optimization beyond strong convexity.

## VISITING EXPERIENCE

- **Sun Yat-Sen University**, Guangzhou, China.
  Host: Prof. Qing Ling. Aug. 2020 – July 2021
- **Michigan State University**, East Lansing, MI.
  Host: Prof. Yuying Xie. July 2019 – Oct. 2019

## PROFESSIONAL SERVICE

**Leadership and Organization**

- Session Co-Chair, Asilomar Conference on Signals, Systems, and Computers, 2024.
- Technical Program Committee, Cross-Community Federated Learning Workshop at MLSys, 2022.

**Reviewer**

- AAAI (2025), AISTATS (2022–2026), NeurIPS (2023–2026), ICLR (2024–2026), ICML (2024–2026).
- IEEE Transactions on Signal Processing and IEEE/ACM Transactions on Networking.
- Journal of Machine Learning Research (JMLR).